A. A. ANGGITA YUNISKARA DEWI

GEOSPATIAL ANALYST | SPATIAL DATA VALIDATION AND REMOTE SENSING

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PROFESSIONAL SUMMARY

BNSP-certified GIS Analyst and Geography graduate of Universitas Indonesia (GPA 3.74) with hands-on experience in spatial data validation, optical and SAR remote sensing, and spatial databases. I validated 5,500+ field survey records at PALYJA, produced 10 environmental and disaster analyses for Kompas.id, and built independent projects ranging from forest carbon estimation with GEDI LiDAR and land clearing monitoring to automated HEC-RAS flood modelling and PostGIS analysis. I work in ArcGIS Pro, QGIS, Google Earth Engine, PostgreSQL/PostGIS, and Python, and document every method and limitation in my portfolio.

CORE COMPETENCIES

Spatial Data Validation and QA/QC | Optical and SAR Remote Sensing | Land Cover and Change Detection | Forest Biomass and Carbon Estimation | Spatial Database Management (PostGIS) | Flood and Hydraulic Modelling (HEC-RAS) | Python and Google Earth Engine Automation | Cartography and Story Maps | WebGIS and Dashboards | Technical Reporting

PROFESSIONAL EXPERIENCE

PT Opac Barata

HR Administrative Staff | *September 2025 – Present*

- Built Excel and Looker Studio dashboards consolidating headcount, inventory, and operational cash flow data across 5 group companies, reducing the time needed to prepare monthly management reports.
- Prepared Monthly Business Reports covering HR KPIs, recruitment progress, workforce movements, onboarding, and turnover to support performance monitoring.
- Managed employee records from onboarding to offboarding in the HRIS and reviewed daily attendance and overtime data against company policy, reporting discrepancies to management.

PT Kompas Media Nusantara

GIS Operator Intern | *September 2024 – December 2024*

- Processed satellite imagery and spatial layers with Google Earth Engine (JavaScript), Google Colab (Python), and GIS software to analyze vegetation loss, land surface temperature, glacier extent, and bathymetry.
- Produced 10 environmental, climate, and disaster analyses for Kompas.id under one-week deadlines, from data selection and accuracy checks to maps, story maps, web apps, and dashboards ready for publication.

PT PAM Lyonnaise Jaya (PALYJA)

GIS QC Operator – Freelance | *October 2024 – December 2024*

- Validated 5,500+ field-surveyed spatial records, checking attribute completeness, geometry, coordinate reference systems, and positional accuracy against satellite imagery, Google Maps, and internal basemaps.
- Worked with field survey teams to trace and fix data discrepancies and documented recurring errors to keep the utility dataset consistent.

GIS AND REMOTE SENSING PROJECTS

Forest Carbon Lost to the Merauke Food Estate, 2023 to 2026

- Sampled pre-clearing forest with 8,577 quality-filtered GEDI L4A footprints, filled the gaps with a Random Forest model on Sentinel-2, Sentinel-1, and DEM data (20% held out for testing) to build a 30 m biomass map, and estimated 11.8 million t CO2 from lost aboveground biomass, cross-checked against GEDI L4B and direct footprint averages.

Quarterly Land Clearing Change Detection, Merauke, 2023 to 2026

- Built cloud-masked Sentinel-2 composites per quarter, detected new clearing with NDVI, bare soil, and water index thresholds against a 2022 vegetated baseline, removed patches under 5 ha, and overlaid about 45,800 ha of clearing with ecosystem type, 2020 forest cover, and forest area status.

Automated Flood Modelling and Machine Learning Emulator, Ciliwung

- Generated and ran 80 HEC-RAS 2D rainfall scenarios entirely from Python, then trained a PCA and Gaussian Process emulator that reproduces flood depth maps with 1.4 cm RMSE in about 4 milliseconds instead of 3 minutes per run.

Flood Extent Mapping with Sentinel-1 SAR, Demak, 2024

- Compared pre- and post-flood Sentinel-1 VV backscatter in Google Earth Engine, applied speckle filtering, change thresholds, and masks for permanent water, steep slopes, and noise, and mapped about 415 ha of estimated inundation.

Flood Exposure and Health Facility Access with PostgreSQL/PostGIS

- Built a PostGIS database with spatial indexes, checked geometry validity and CRS, and used buffer, spatial join, intersection, and nearest-neighbour queries on practice data to flag villages with over 20% flood exposure and no health facility within 1 km.

Multi-Source Deforestation Screening and WebGIS, Bengkalis, Riau

- Combined the JRC 2020 forest baseline with Hansen, RADD radar alerts, and Ministry of Forestry data on a common grid to screen 284,918 ha of oil palm, and built a Leaflet app where users upload GeoJSON, shapefile, or CSV plots and download per-plot results.

Publications

- Authored four published articles, three as first author, on environmental and climate risk topics, including air quality, flood hazard mapping, and temperature distribution, in Jurnal Geografi, Lingkungan dan Kesehatan (2024) and Majalah Geospasial (2023, 2024).

EDUCATION

University of Indonesia

Bachelor of Geography (GPA: 3.74/4.00) | *August 2021 – July 2025*

- Thesis: Dynamics of Air Quality in Denpasar City and Its Surrounding Areas from 2019 to 2024. Analyzed NO2 and SO2 from Sentinel-5P and DLHK ground data and tested land-cover drivers with Pearson correlation.

TECHNICAL TRAINING

GIS Training – Geoaccess | *August 2026 – September 2026*
Practical training in ArcGIS Pro, georeferencing, spatial and attribute queries, vector and raster processing, geoprocessing, ModelBuilder, cartography, metadata, and spatial data quality control.

Integrating AI for Spatial Data Analysis (1-on-1 Learning) | *October 2026*
Using AI to support spatial analysis, including connecting QGIS and Google Earth Engine to an AI assistant.

Geotechnical Engineering for Open Pit Mining | *October 2026*
Rock mass characterization, slope design and stability analysis, geotechnical monitoring with Trigger Action Response Plans, and numerical modelling for open pit mining.

Environment Officer Online Training – Exagama | *October 2026*
AMDAL, UKL-UPL, SPPL, PROPER compliance, hazardous waste (B3) management, and environmental auditing.

ESG Career Workshops – Farisphere | *October 2026*
ESG entry points including climate and carbon, ESG risk, supply chain, sustainable finance, and reporting frameworks, and translating GIS experience into ESG work.

Data Analyst Bootcamp – HariSenin | *April 2026 – June 2026*
Excel-based analysis, SQL, Python, statistics, and dashboards in Power BI, Tableau, and Looker Studio.

Python for Data Science Bootcamp – PT Hacktivate Teknologi Indonesia | *August 2023 – December 2023*
Data preprocessing, analysis, and visualization using Python, Pandas, NumPy, Matplotlib, and Seaborn.

CERTIFICATIONS

- **Geographic Information Systems Analyst –** BNSP | September 2026 – September 2029

ACHIEVEMENTS

- **Silver Medal** – Diponegoro Science Competition, 2024 | Website Invention for Stunting Prevention in NTT
- **Bronze Medal** – The 9th KIME on Ideas Competition, 2023 | Spatial Study of Landslide-Prone Areas in Nglipar, Gunungkidul

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TECHNICAL SKILLS

GIS & Remote Sensing: ArcGIS Pro, QGIS, Global Mapper, Google Earth Engine, Sentinel-1 SAR, Sentinel-2, Landsat, GEDI, DEMNAS

Data & Modelling: Python (Pandas, NumPy, geemap), PostgreSQL/PostGIS, SQL, GDAL, HEC-RAS, ModelBuilder, SPSS

WebGIS & Reporting: Leaflet, MapLibre GL JS, JavaScript, ArcGIS StoryMaps, Looker Studio, Power BI, Microsoft Excel